RATAN TATA MAHARASHTRA STATE SKILLS UNIVERSITY

School of Science Engineering and Technology Skills

Department of Engineering

Subject –DevOps and Automation Lab

Lab Assignment 9

Aim:

Create a basic Jenkins Pipeline using Jenkinsfile to build a Maven project.

Learning Objectives:

Create a Jenkins Pipeline that:

- Pulls code from GitHub
- Builds a **Maven project** using Jenkinsfile

Steps:

STEP 1: Create a Sample Maven Project

You can either:

- Use your own project
- OR create a simple one

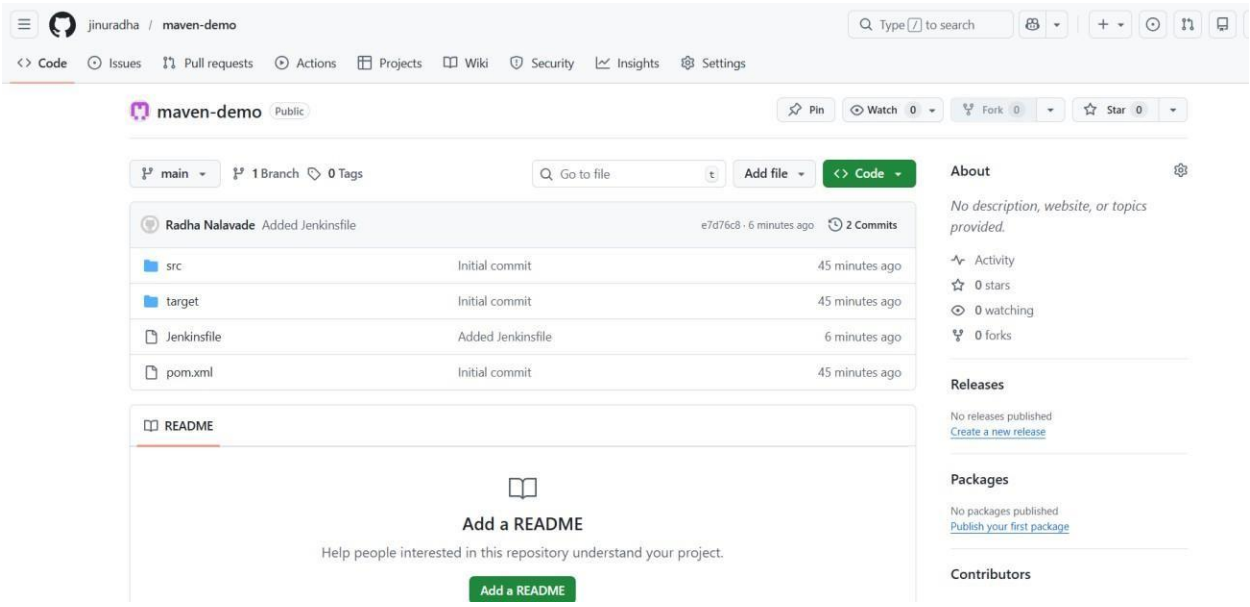
Project structure:

```
my-maven-project/
├── pom.xml
└── src/
    └── main/java/App.java
```

STEP 2: Push to GitHub

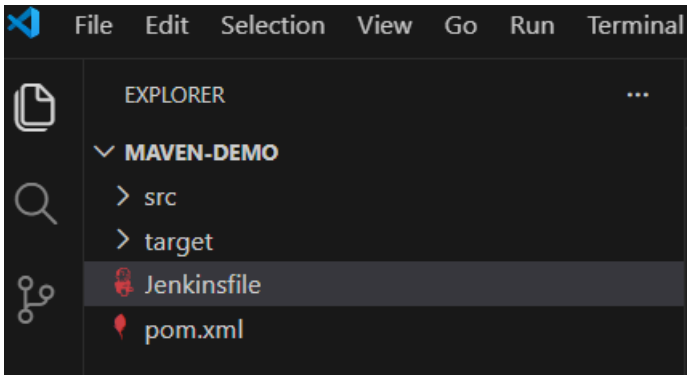
Go to GitHub

1. Create repo: *maven-jenkins-demo*
2. Upload:
    - pom.xml
    - src/ folder



STEP 3: Create Jenkinsfile

Inside your project, create file: *Jenkinsfile*



Add this code in this file:

```
pipeline {
  agent any

  tools {
    maven 'Maven'
  }

  stages {
    stage('Clone') {
      steps {
        git    'https://github.com/your-username/maven-jenkins-demo.git'
      }
    }

    stage('Build') {
      steps {
        bat 'mvn clean compile'
      }
    }

    stage('Test') {
      steps {
        bat 'mvn test'
      }
    }

    stage('Package') {
      steps {
        bat 'mvn package'
      }
    }
  }
}
```

Important:

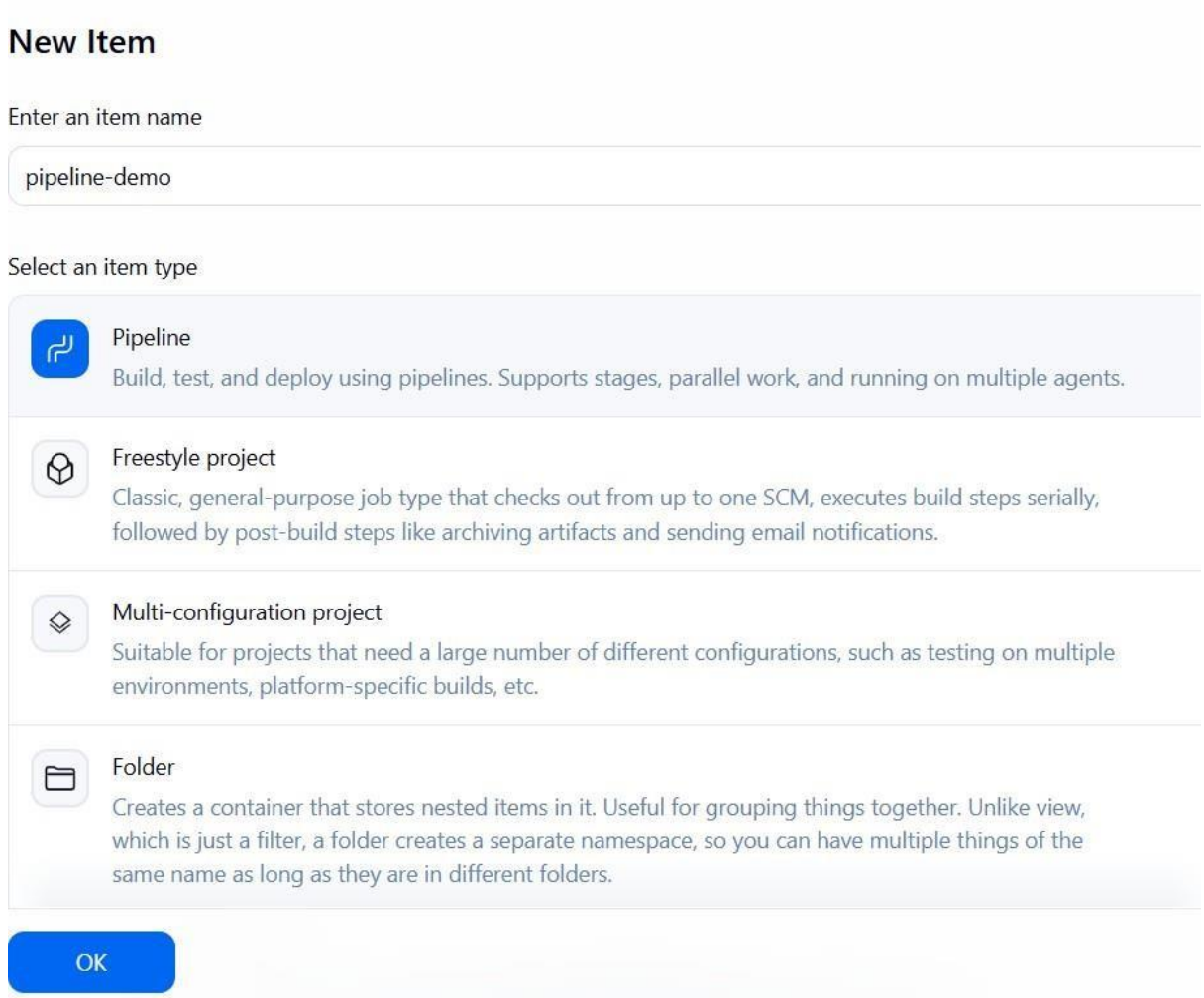
Make sure you add your git url with your username.

STEP 4: Configure Maven in Jenkins

1. Go to Jenkins: *Manage Jenkins → Global Tool Configuration Scroll to* ***Maven***
2. Click: *Add Maven*
3. Fill: *Name: Maven*
4. Tick: *Install automatically*
5. Select *latest version (or default)*

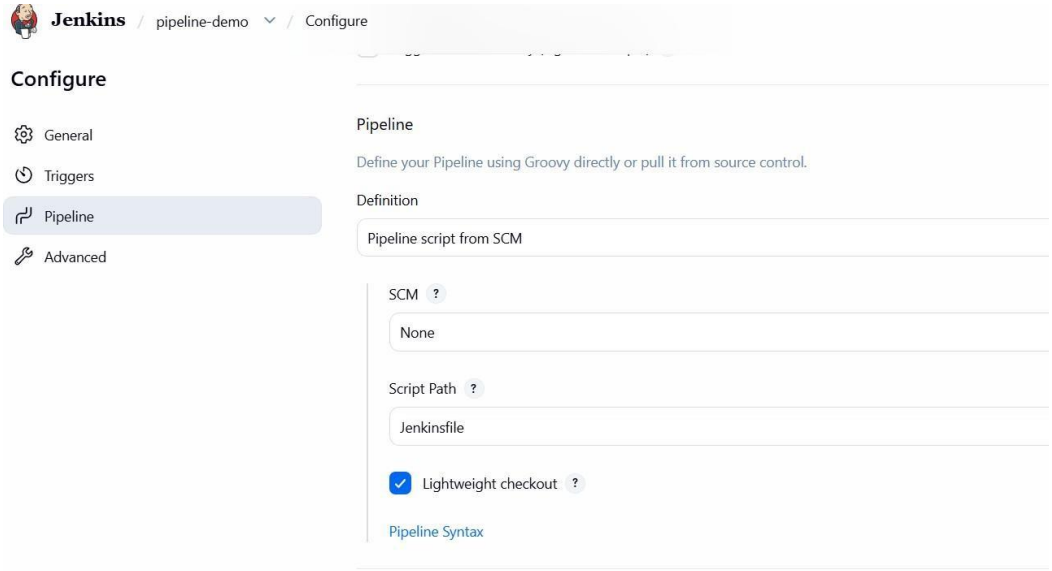
STEP 5: Create Pipeline Job

1. Click: *New Item*
2. Name: *pipeline-demo*
3. Select: *Pipeline*



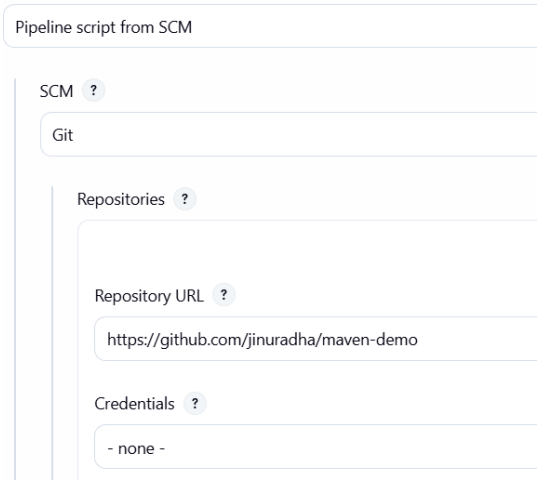
Configure Pipeline

Scroll to: *Pipeline → Definition Select: Pipeline script from SCM*



SCM Settings:

- SCM: *Git*
- Repo URL: https://github.com/your-username/maven-jenkins-demo.git
- Script Path: *Jenkinsfile*



STEP 6: Run Pipeline

1. Click: *Build Now*
2. Check console output.